

Problem 5. Let ABC be a triangle with $AB \neq AC$ and circumcircle Ω . Its incircle has centre I and touches the side BC at point D . Line AI intersects Ω at M again, and DM intersects Ω again at P .

Prove that $\angle API = 90^\circ$.

Problem 6. Let ABC be a triangle with points D and E on segments AB and AC , respectively, such that $DB = BC = CE$. Let the lines CD and BE meet at F .

Prove that the incentre I of triangle ABC , the orthocentre H of triangle DEF and the midpoint M of the arc BAC of the circumcircle of triangle ABC are collinear.

Problem 7. Two points P and Q lying on side BC of triangle ABC and their distance from the midpoint of BC are equal. The perpendiculars from P and Q to BC intersect AC and AB at E and F , respectively. Let M be point of intersection PF and EQ and let H_1 and H_2 be the orthocentres of triangles BFP and CEQ , respectively.

Prove that $AM \perp H_1H_2$.

Problem 8. Given two fixed circles ω_1, ω_2 , let $\mathcal{S}$ denote all $\triangle ABC$ satisfying the following conditions:

- ω_1 is the circumcircle of $\triangle ABC$,
- ω_2 is the A -excrcle of $\triangle ABC$ which touches BC, CA, AB at D, E, F , respectively, and
- $\mathcal{S}$ is non-empty.

Prove that the centroid of $\triangle DEF$ is a fixed point.